

Criterion B: Developing ideas

i. Design specifications

My specifications come straight from my Criterion A research and are organised using ACCESSFMM. For each one I explain what it is, why I chose it (linked to my primary and secondary research), and who will test it, when and how. Every specification has a measurable success target, so there is a clear line between met and not met.

Where a test uses a rating, the client or peers score from 0 to 5 (0 = very bad, 1 = poor, 2 = satisfactory, 3 = good, 4 = very good, 5 = excellent). A rated specification is met when the average is 4/5 or higher and its measurable target is also hit. The feedback I refer to is summarised in Appendix B1.

ACCESSFMM	The Specification (The Requirement)	Explanation (What it is)	Justification (Why it was chosen)	Testing Method (Who, When, How)
Aesthetics	Calm and consistent look. The app will use no more than three main colours plus one neutral text colour, with rounded corners and the same style on every screen.	Warm ivory and sand backgrounds with one muted sage green accent and a warm near-black for text. Cards and buttons have rounded corners (12 to 16 px). There is no red anywhere, not even on the help button. Every screen shares the same colours, font and corner style, so the app feels like one calm space rather than separate pages.	In my Criterion A analysis (Aiii), Wysa's teal and cream branding is deliberately warm and non-clinical, the opposite of the intense red used by hotlines (Wysa, n.d.). A student who opens the app mid-class is already stressed, so the screen should lower that, not look like an emergency service. Pause takes the same warm, non-clinical idea in its own beige and sage colours. In my peer survey, 14 of 20 students said apps that look like a hospital or clinic put them off (Appendix B1).	Who: 6 peers (at least one from each year, MYP4 to DP2). When: on the B.ii wireframes, and again in Criterion D. How: they look at every screen and rate 'calm' and 'consistent' from 0 to 5. I also count the colours in the app's theme settings. Success: 3 main colours or fewer + 1 neutral, 0 reds, and an average of 4/5 or higher on both questions.
Cost	Free to build and free to use. The app will cost 0 OMR to build and host, and students will never pay, sign up or see adverts.	I build and host it on free tiers only. The check-in, suggestions, Help screen and counsellor button all work without payment or an account.	Both products I analysed in Aiii put features behind paid plans (Wysa, n.d.; Daylio, n.d.). A paywall in front of help would block the students who need it most. My related concept of resources frames this project as stretching limited counsellor time further, not selling something.	Who: me (the designer). When: at the end of Criterion C. How: I open the app in a private browser window on a device that has never used it and go through the full flow. I then check my resource log and the platform's billing page. Success: 0 payment prompts, 0 sign-up prompts, 0 adverts and 0 OMR spent.

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Customer / Client	Fits both MYP and DP students. Students from MYP4 to DP2 will be able to describe what is stressing them with tags that cover both MYP and DP pressures.	Two groups of tags. 'How I feel' (12): anxious, stressed, overwhelmed, frustrated, low, tired, then angry, lonely, embarrassed, restless, unmotivated and 'something else' behind a 'More' chip. 'What's happening' (10): test or exam, IA or deadline, in class, presentation, homework, friends or people, then home, sleep, social media and 'something else' behind 'More'. Tapping a feeling offers an optional 'Tell us more' sheet with an exact word (explained in brackets), how strong and how long. Every tag is 3 words or fewer and uses words a 14-year-old already uses.	Criterion A treats MYP4 and DP2 stress as different (related concept: perspective). An MYP4 student starting the Personal Project is not under the same load as a DP2 student with IA deadlines. My primary research found that stress during difficult exams reduced students' mental capacity and lowered their grades (Aiv). In my peer survey, exams (16/20) and deadlines (13/20) were the two most common triggers (Appendix B1).	Who: 3 MYP4 to MYP5 students and 3 DP1 to DP2 students. When: during the B.iii feedback session. How: each student is asked: 'Could you describe your last stressful school moment using only these tags?' and rates the fit from 0 to 5. I average MYP and DP separately so one group cannot hide the other. I also count the words in every tag. Success: an average of 4/5 or higher in both groups, and 10/10 tags are 3 words or fewer.
Customer / Client	Allowed in class. The app will work silently during a lesson, and the counsellor will approve all safety wording before it is used.	No sounds, vibration, push notifications, or animation longer than 5 seconds. The counsellor reviews and approves the Help screen text, contact details and the 'you're not alone' message.	Criterion A names teachers and counsellors as my secondary client: they decide whether the tool is allowed in class and receive any escalation. Stress happens mid-class (Ai), so the app must work there without drawing attention to the student. The teacher I spoke to said they would allow it only if it is completely silent (Appendix B1). I took the 5-second limit from WCAG 2.2, which says anything that moves by itself for more than 5 seconds needs a pause control (W3C, 2024, SC 2.2.2). Keeping every animation shorter than that is simpler than adding another button.	Who: me, 1 teacher and the counsellor. When: on the Criterion C prototype. How: I do a full check-in on 3 devices with the volume at maximum. The teacher rates 'I would allow this in my lesson' from 0 to 5, and the counsellor signs off the wording. Success: 0 sounds, vibrations or notification requests and 0 animations over 5 s on 3/3 devices, a teacher rating of 4/5 or higher, and counsellor sign-off.
Environment	Works fast on school Wi-Fi. The app will open in 10 seconds or less on school Wi-Fi and show a suggestion within 2 seconds of submitting.	It is used in classrooms, the library and between exams, on phones and school laptops over school Wi-Fi. The matching runs on the device itself, so there is no server to wait for.	Criterion A argues the solution must exist where the stress occurs, not before or after it. Bentley et al. (2022) gave 43 high school students a breathing technique to practise alone, and compliance fell from 73% to 47% over five weeks. In Criterion A I argued the delivery failed because nothing reached students in the moment. A slow app would repeat that mistake.	Who: me. When: during Criterion C and in final testing (Criterion D). How: I load the app 3 times on each of 3 school devices with the cache cleared and time it with a stopwatch. Then I time 10 submissions. Success: 9/9 loads in 10 s or less, and 10/10 suggestions in 2 s or less.

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Environment	Lightweight text. The app will download 0 KB of font files by using system fonts. If a web font is ever needed (e.g. for Arabic), the limit is 100 KB in total.	It uses fonts already installed on the device, one font family, and no autoplay video or decorative images. Any future web font uses the WOFF2 format.	jacopo (2025) explains that fonts are data files downloaded with every page, that one custom web font can add hundreds of kilobytes, and recommends system fonts, one family and WOFF2 to cut data transfer and energy use. A 100 KB cap sits below what one custom family can add. Less data also helps the 10-second load time above.	Who: me and 3 peers. When: during Criterion C. How: I open the browser's Network tab, filter by 'Font', reload and add up the sizes. Then 3 peers read a suggestion card aloud from a phone held at arm's length. Success: 0 KB of fonts (100 KB or less if a web font is added), and 3/3 peers read the card without errors.
Safety	Counsellor button on every screen. A 'Talk to the counsellor' button will be visible on every screen, without scrolling, in the same place each time.	It sits in a fixed bar at the top and is at least 44 × 44 px. It opens a Help screen with the counsellor's name, room, how to drop in or book, and the emergency number.	In Aiii, Wysa showed that a default, always-visible safety response works better than one the user has to look for (Wysa, n.d.). Criterion A also states that the app must include a visible, unconditional link to the counsellor so it never replaces professional support. WCAG 2.2 says help options repeated across pages should stay in the same order (World Wide Web Consortium [W3C], 2024, SC 3.2.6).	Who: me and 5 peers. When: on the B.ii wireframes, and in Criterion D. How: I screenshot every screen at 320 px and 1366 px without scrolling. Each peer starts on a random screen and is asked: 'Find how to contact the counsellor.' I time them. Success: the button is in the same place on 100% of screenshots, and every peer finds it in 5 seconds or less.
Safety	A 'you're not alone' line for everyone. The suggestion screen and Help screen will show the same short support message, worded with safe-messaging guidance.	Example: 'If things feel like too much right now, you don't have to handle it alone. The school counsellor is here for you.' It is never triggered by what someone types. Every user sees it every time.	Wysa's AI detection caught 82% of crisis instances (Aiii), which means it missed almost one in five. A line shown to everyone cannot miss anyone. In Aii I planned to use safe-messaging guidance so the wording is tested rather than invented. I will check it against Orygen's #chatsafe guidelines, which were written with young people (Orygen, n.d.), and the counsellor has the final say.	Who: the counsellor and 6 peers. When: before building (Criterion C), and in Criterion D. How: the counsellor approves or edits the wording. Peers rate it from 0 (cold or clinical) to 5 (supportive). Success: counsellor sign-off, and an average peer rating of 4/5 or higher.

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Safety	Private by design. Check-in entries will stay only on the student's own device, and the app will never ask for a name, email or student ID.	Entries are kept in the browser's own storage. Saving history is off by default because school devices are shared. A 'Delete my entries' button clears everything.	Criterion A (Ai) found that some students want their problems kept from their parents, while the counsellor is obliged to inform parents. If entries sat on a server, students would stop being honest in the check-in. Daylio keeps entries off its own servers (Aiii), which shows a mood log does not need a central database.	Who: me. When: during Criterion C and in Criterion D. How: with the browser's Network tab open, I do 5 check-ins containing the test word 'TESTPRIVACY' and search every request for it. Then I check the stored data, press 'Delete my entries' and check again. Success: 0 requests contain the test word, and storage is empty after deleting.
Safety	Nothing unsafe for minors. The app will contain no social features, no AI reading of what users type, and no supplement or medication content.	No posts, comments, sharing or chatting with other users. No chatbot or crisis detection. Every suggestion is a non-drug technique: slow breathing, grounding, CBT-style reframing, a short movement or break, or reaching out to someone.	Wysa's detection missed 18% of crisis instances (Aiii), and Woebot shut down its app in 2025 (HLTH, 2025) despite having trial evidence. Reliable detection is far beyond a 20-hour build. A peer feature would put one minor's disclosure in front of others with no trained moderator. My users are 14 to 18, so only non-drug techniques are appropriate. The counsellor asked that nothing reads or analyses what students write (Appendix B1).	Who: me and the counsellor. When: at the end of Criterion C. How: I check every screen against a list of banned features and search all text for banned words (supplement, nootropic, medication, pill, dose, mg). The counsellor reviews the result. Success: 0 banned features, 0 banned words, and the counsellor confirms.
Size	Fits every school screen. The layout will fit screens from 320 px to 1366 px wide with no sideways scrolling and nothing overlapping.	This covers small phones up to school laptops.	My design brief (Aiv) requires support for all relevant platforms, and students use their own phones as well as school laptops. WCAG 2.2 sets 320 px as the width content should fit without scrolling sideways (W3C, 2024, SC 1.4.10).	Who: me. When: in B.iv and Criterion C. How: I use the browser's device toolbar to view every screen at 320, 375, 768 and 1366 px. Success: 0 sideways scrolling and 0 overlaps at all 4 widths.
Size	Easy to tap and read. Every button will be at least 44 × 44 px, body text at least 16 px, and text contrast at least 4.5:1.	Targets have at least 8 px of space between them. Headings are at least 20 px and line height is 1.5. Mood buttons have word labels, so colour is never the only signal.	WCAG 2.2 sets 44 × 44 px as the enhanced target size (SC 2.5.5), 4.5:1 as the minimum contrast for normal text (SC 1.4.3), and says colour must not be the only signal (SC 1.4.1) (W3C, 2024). 16 px is the default text size in browsers, so nothing is smaller than what students read every day. A mis-tap costs the seconds this app is meant to save.	Who: me and 5 peers. When: during Criterion C. How: I check every text and background pair with the WebAIM Contrast Checker and measure sizes with the inspect tool. Each peer taps every button once while I count mis-taps. Success: 100% of pairs pass, 100% of targets are 44 × 44 px or larger, 0 text below 16 px, and 1 mis-tap or fewer in total.

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Form and Function	Hybrid check-in on one screen. One screen will hold a 5-point mood scale, the feeling and situation tags and an optional text box of up to 280 characters. The optional detail sheet opens over this screen, not as a new one.	Choosing a mood is required; tags and text are optional. One button, 'Show me something that might help', submits the check-in.	Daylio shows that a mood-then-tags picker is fast (Aiii), but on its own it gives nothing back. My brief (Aiv) lists 'acts as a mental journal' as a key requirement, so the text box lets students say what the tags miss. It is optional so it never slows anyone down. 12 of 20 peers said they would want somewhere to write 'when it's something specific' (Appendix B1).	Who: me. When: on the B.ii wireframes and in Criterion C. How: I screenshot the screen at 320 px and 1366 px and tick off the four parts. Then I submit 5 times with only a mood chosen. Success: all 4 parts on one screen at both widths, and 5/5 submissions work with the text box empty.
Form and Function	A matched coping suggestion. After a check-in, the app will show one suggestion from a library of at least 12, chosen by a fixed mapping table.	Each card shows the technique's name, 2 to 4 short steps, and a 'Try a different one' button. The library covers 5 technique types. The full mapping table is in B.iv. No AI is used.	This closes the gap I found in Daylio, which records a mood but never responds (Aiii). Bentley et al. (2022) suggest the delivery, not the technique, was what failed, so the suggestion has to appear at the moment of the check-in. A fixed table means I can check every possible output before anyone uses the app.	Who: me and the counsellor. When: during Criterion C. How: I enter 5 different input sets (listed in B.iv) and compare each result with the mapping table. The counsellor rates how suitable the suggestions are from 0 to 5. Success: at least 3 different suggestions, 5/5 match the table, and a counsellor rating of 4/5 or higher.
Form and Function	Fast enough to use under stress. A student will reach a suggestion in 3 taps or fewer and 15 seconds or less from the check-in screen.	The shortest path is: choose a mood, choose one tag, submit.	My primary research found that exam stress reduced students' mental capacity and lowered their grades (Aiv). Arnsten (2009) explains a likely reason: even mild, uncontrollable stress can quickly weaken prefrontal cortex abilities such as working memory and controlling attention. A stressed student has little capacity to spare on an app, so every extra step is a reason to give up. In my survey, 15 of 20 students said they would stop using an app that took more than about 30 seconds (Appendix B1).	Who: 5 peers across MYP4 to DP2. When: on the Criterion C prototype and in Criterion D. How: each gets a scenario card: 'You just walked out of a hard test and feel anxious. Use the app to find something that helps.' I time them, count taps, and ask them to rate ease of use from 0 to 5. Success: median time of 15 s or less, every tester uses 3 taps or fewer, and an average ease rating of 4/5 or higher.

ACCESSFMM	The Specification (The Requirement)	Explanation (What it is)	Justification (Why it was chosen)	Testing Method (Who, When, How)
Form and Function	Honest about the evidence. Every suggestion will show a one-line reason it might help, an evidence-strength label and its source.	Strong evidence: backed by systematic reviews or clinical guidelines. Some evidence: backed by trials, but with small or mixed results, or not tested with teenagers. Worth a try: low risk and widely recommended, with little direct research. No card uses the words 'cure', 'fix', 'treat' or 'guaranteed'.	My brief (Aiv) requires evidence-based strategies, but evidence-based does not mean proven for every student. Bentley et al. (2022) found no statistically significant change in students' measured stress or anxiety after their 5-week breathing pilot. A label shows students how strong the support really is, so the app never promises more than the research shows.	Who: me and the counsellor. When: at the end of Criterion C. How: I check every card for all three parts and search for the banned words. The counsellor reviews the labels on 3 randomly chosen cards. Success: 100% of cards are complete, 0 banned words, and the counsellor agrees with 3/3 labels.
Manufacturing	Built in a browser in about 20 hours. The app will be a web app built with free AI-assisted tools, with no login before the check-in.	I plan the screens in Google Stitch, build the app in Google AI Studio, and publish it as a simple website on free hosting. It runs in Chrome, Safari or Edge on iOS, Android, Windows, macOS and ChromeOS without installing anything. The finished app contains no AI. The AI tools only help me write the code, and every suggestion comes from my fixed mapping table.	In Aii I planned to find a vibe-coding platform that can build this in 20 hours. My brief (Aiv) requires support for all platforms, and building a separate native app for each one would not fit the time. A login screen would add taps before help and make students identify themselves, which breaks the privacy specification.	Who: me. When: throughout Criterion C. How: I keep a time log against my plan. At the end I complete the full flow on 3 school devices with different operating systems and 1 phone. Success: built in 20 hours or less, 4/4 devices complete the flow, and 0 login screens before the check-in.
Manufacturing	Easy to extend later. All text and the mapping table will live in one editable data file, so new suggestions or an Arabic version need no rebuild.	Labels, suggestions, the support message and the mapping table sit in one table instead of being typed into each screen. The layout also works right-to-left.	My related concept of resources is about stretching counsellor time to more moments, and reaching more students means more suggestions and, in Oman, Arabic. The counsellor said some MYP students would be more comfortable in Arabic (Appendix B1). I also plan to keep developing the app after this unit.	Who: me. When: at the end of Criterion C. How: I time myself adding 1 new suggestion with its mapping row and translating 3 labels. Then I switch the layout to right-to-left and check every screen at 320 px. Success: done in 15 minutes or less, 0 screens rebuilt, and 0 overlaps in right-to-left.
Materials	Free, licensed and traceable. The app will use only free tools, system fonts, free-to-use icons and text I write myself from sources in my reference list.	I keep an asset register listing every icon, tool and piece of content, where it came from, and its licence.	For a digital product, the materials are its software and content. Using only free resources keeps the cost at 0 OMR, and recording a source for every piece of content is what makes the evidence labels honest.	Who: me. When: at the end of Criterion C. How: I check the asset register against everything visible in the app. Success: 100% of assets have a source and a licence recorded.

Appendix B1: Client and peer feedback used in my specifications

Peer survey	Anonymous Google Form, 20 students (5 each from MYP4, MYP5, DP1 and DP2), run during form time. Most common stress triggers (students could pick more than one): tests or exams 16/20, IA or deadlines 13/20, presentations 7/20, friends 6/20, home 4/20. 15/20 would stop using an app that took more than about 30 seconds. 12/20 wanted somewhere to write freely. 14/20 said apps that look like a hospital or clinic put them off.
Counsellor interview	20-minute interview with the school counsellor. Students most often come to see her after mock exams and near IA deadlines. She does not want anything that reads or analyses what students write, and wants to approve every piece of help wording and contact detail. She said some MYP students would be more comfortable in Arabic.
Teacher conversation	Short conversation with one MYP teacher. They would allow the app in a lesson only if it is completely silent and quick: 'If it makes a sound, I'd have to ban it.'

Appendix B2: Questions I asked

Peer survey questions	1. Which year are you in? (MYP4 / MYP5 / DP1 / DP2) 2. When do you feel most stressed at school? (tick all: tests or exams, IA or deadlines, presentations, friends, home, other) 3. If an app took longer than ____ to help you, you'd stop using it. (10 s / 30 s / 1 min / 2 min+) 4. Would you want a place to write freely, or only tap options? (write / tap / both) 5. Rate how much an app that looks like a hospital or clinic would put you off (0 to 5).
Counsellor interview questions	1. When do students most often come to you, and about what? 2. What should an app like this never do? 3. What should the Help screen say, and which contact details should it show? 4. Which coping techniques are you comfortable with the app suggesting? 5. Would an Arabic version help students here?
Teacher question	'Would you allow a student to open this during your lesson? What would make you say no?'

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